ECE153A Final
Autumn Quarter
December 10, 2001
UC Santa Barbara
Total 120 pts

Name:

I. (30 pts) **True or False questions**. Indicate whether each of the following statement is *TRUE* or *FALSE*. If you choose *FALSE*, you must explain the reason(s) or NO credit is given.

1 (Power Management): Using a very large T_{TO} is safe and cannot do worse (in terms of saving power) than doing no power management at all.

2 (Process Scheduling): Suppose a system employs a time-sliced (quantum) and round-robin scheduling policy. Having a large time-slice (quantum) improves the average response time of the processes in the system. (Response time is defined as the time between when a process is created and when the process is placed in the running state for the first time.)

3 (Power Management): Suppose $P_{off,on} = P_{on}$ and $T_{off,on} > 0$. To conserve power, the system should place a device in the sleep mode (i.e., the soft-off mode) whenever the device is idle.

4 (Device Drivers): Character devices differ from block devices only in the way that they do IO. Whereas character devices read and write data to and from the system buffer cache, block devices transfer data to and from user data buffers.

5 (Device Drivers): In the middle of servicing an IO-completion interrupt, the kernel may be interrupted by another IO-completion interrupt that has a higher interrupt priority.

6 (Storage Management): Since the price of DRAM becomes cheaper and cheaper every year. Eventually, a computer's storage can be entirely DRAM-based, and magnetic disks can be entirely eliminated.

II. (30 pts) For each question, choose the letter (a), (b), (c), (d), or (e) as your answer. Score = (5× number right) - number wrong, so random guessing nets you nothing on the average. If you write down the steps that arrive at your solution to a problem, you may receive partial credit.

1: (Power Management) Suppose that your power manager predicts that the next device idle duration will be shorter than T_{Be}. So, it does not place the device in the soft-off state when the device is idle. But the next device actual idle time is longer than T_{Be}. What is the power consumption status of the device?
(a) The device saves power (b) The device neither saves nor wastes power (c) The device wastes an opportunity to save power (d) indecisive.

2: (Power Management) Suppose a device is on and it has the following parameter values: $T_{off,on}$ = 60 seconds, $P_{off,on}$ = 100 watts, P_{off} = 30 watts, P_{on} = 60 watts. If the next T_{idle} is predicted as 120 seconds with 100% certainty. The best power management decision is
(a) Transition the device to the off state immediately.
(b) Transition the device to the off state if the device is idle for 10 seconds.
(c) Transition the device to the off state if the device is idle for 60 seconds.
(d) Transition the device to the off state if the device is idle for 140 seconds.
(e) Leave the device on.

3: (Real-time Scheduling) You are given the following four periodic tasks. Identify a feasible real-time schedule that maximizes the total importance factor ($\sum I_i$).

Task i	E_i (ms)	T_i (ms)	I_i (importance factor)
1	12.5	50	5
2	25	100	3
3	40	200	6
4	100	400	2
5	200	800	4

(a) All five tasks (b) Task 1, 3 and 5 (c) Tasks 1, 2, 3 and 4 (d) Task 1, 2, 3 and 5 (e) Tasks 1, 2, 4 and 5.

(Device Drivers) Suppose a disk drive has a uniform recording density on all tracks and it has the following performance parameters. Answer the following two questions. (Hint: Because of the uniform recording density, a longer track has a higher recording capacity.)

Parameter Name	*Value*
Disk Capacity	*50 GBytes*
Number of cylinders, CYL	*10,000*
Number of double-side disk platters	*10*
Min. Transfer Rate TR	*100 Mbps*
Max. Transfer Rate TR	*140 Mbps*
Min. Seek Time	*3 milliseconds*
Max. Seek Time	*15 milliseconds*

4: What is the capacity of the disk's inner-most cylinder? (a) 2M Bytes (b) 3.33M Bytes (c) 4.17M Bytes (d) 6.67M Bytes (e) 6.86M Bytes.

5 What is the minimum revolutional speed of the disk in RPM? (Hint: Consider double-side platters.) (a) 1,000 (b) 3,600 (c) 4,800 (d) 7,200 (e) 10,000.

6: Which of the following are (is) the desirable properties of an Interrupt Service Routine?

I: The routine must not be interruptible .

II: The routine should defer its non-critical tasks to lower interrupt-priority levels as much as possible.

III: The routine must not sleep/block.

(a) I and III (b) II and III (c) I, II, and III (d) III (e) II and III.

III. (30 pts) Short Answers.

1 (Real-time Scheduling): The exact test always finds whether a given set of tasks are schedulable but the RMS test can be indecisive. Why a system may still prefer using the RMS test over the exact test to do its schedulability test?

2 (Device Drivers): Define *DMA* and enumerate two data-transfer situations where DMA cannot be helpful for improving IO efficiency.

3 (Device Drivers): Explain why a block device is the desired choice for managing disks and tapes over a character device.

4 (Memory Management): Describe what is an *Inverted Page Table* and what is its primary purpose and advantages over the traditional virtual to physical page table.

5 (Period Transformation): Consider two periodic tasks given in the table below. They are not schedulable based on the exact test. Assume a context switch is free. Perform a period transformation on the second task so that the tasks are schedulable. There are more than one solution to this problem. Please state clearly what is your new T_1 and T_2 and how you arrive at your answer, or no credit will be given. (Hint: There is a simple solution if you know the "shot-cut.")

Task i	E_i (ms)	T_i (ms)
1	60	100
2	60	150

IV. (15 pts). Consider four periodic tasks given in the table. Answer the following two questions (must show all work or no credit is given):

1) According to the RMS test, can all these four tasks meet their deadlines? (5 pts)

2) Using the Exact Schedulability Test to find a set of schedulable tasks that maximizes CPU utilization (10 pts).

Task i	E_i (ms)	T_i (ms)
1	10	40
2	20	60
3	60	120
4	80	200

V. (15 pts) Suppose a system employs a Multi-Level Feedback policy with the following three scheduling rules:

1. There are three service queues in the system, each uses a round-robin scheduling policy. The first queue (the highest priority queue) has a quantum of 30 ms. The second queue has a quantum of 60 ms. The third queue (the lowest priority queue) has a quantum of 100 ms. A higher priority process always preempts a lower priority process. A preempted process is always given its full quantum back. A context switch takes 10 ms.

2. A newly arrived process is placed in the second queue.

3. A process' priority is demoted after a quantum-end and is promoted after an IO completion.

Suppose two processes, P_1 and P_2, arrive at time 0, and 50 ms, respectively, and they have the following CPU and IO bursts:

- P_1: CPU burst: 200 ms (only one burst).
- P_2: CPU burst: 40 ms; IO burst 30 ms; CPU burst: 100 ms (total three bursts).

Answer the following two questions:

(b) What is the response time of each process? (5 pts)

(b) What is the turnaround time of each process (must show all work or no credit is given)? (10 pts)